

kin_ – Dancing with Believable Avatars

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Figure 1: *kin_*: Picture of the Artist and the Performing Avatar.

ABSTRACT

Reflecting on the challenges and potentials that Mixed Reality media present to produce digital performance art, we present the AR artwork *kin_*. The piece is opening the question on how to transfer a real live performative experience into augmented reality as well as the question of owning and maintaining agency within an artistic fabric. This is explored with a focus on the interaction of different types of agents, using artistic research at the intersection of art, dance, and technology.

Index terms: Contemporary Dance, Agency, Avatar, Augmented Reality, Performance, Extended Reality, Ethics.

1 DESCRIPTION OF ARTWORK

How can an art performance be transferred into digital context and what happens when the audience meets performers in digital form? There are many questions that unfold from this initial one, bringing into conversation the rich theoretical background of performance art on the one hand and the theory of digital media, especially Mixed Reality, on the other [1,3,4].

With all due acknowledgment of this uneasy and flexible terrain, for the argumentation purposes of this paper we agree to the definition of performance art as practice that involves a person or persons undertaking an action or actions within a particular time

frame in a particular space, central to its execution being the live presence of the artist and the real actions of his/her body to create and present an ephemeral art experience for an audience. The creation in the moment of performing has been crucial to performance art itself since its beginnings. Liveness has also an important part in the experience of the piece for the viewer. As an art form that traditionally relies on its physical immediacy, bound to its mediums of body, time and space, performance art seems to be multiply challenged when coupled with the medium of mixed reality, which blends the physical and digital world into a new form of interaction and co-existence.

The mentioned live presence or real actions are in this context challenged or, more precisely, we are challenged to reconsider the meaning of performance art in the medium that possibly disintegrates its very foundations, molding it into a new digital form that is yet to be defined. While the first part of our initial question – how an art performance can be transferred into digital context – tackles the aspects of technological possibilities and practical choices, the second one – what happens when the audience meets performers in digital form – invites a more philosophical discussion on the experience of human-avatar encounters within a digital medium and its meaning in performance creation. We argue that with Mixed Reality it might be possible to attempt the idea of liveness within a digitized dance piece without losing the artistic and bodily quality to technology. The discussion on these questions is being built upon the example of the performative art piece *kin_* and we refer to the video for deeper insights on the piece.

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Figure 2: *kin_*: The AR Experience *kin_* in an Indoor Setting.

The interactive Mixed Reality dance performance *kin_* is a dance piece with one up to three moving avatars and audience, developed for Augmented Reality on iOS devices. Several derivatives, such as a full version for the Apple Vision Pro and a semi-interactive Web AR version have been developed. Additionally, a second part, the interactive dance piece *proxy* has been premiered and shown worldwide [2].

The first part of the piece invites the user to get to know three avatars, their space, functionality, and behaviours, such as introducing choreographic material and setting the scene to slowly immersing the user into a Mixed Reality setting. The second part consists of a fast choreographed trio of all three avatars dancing on relations and postures. Meanwhile the avatars do not take notice of the spectator in the second part, they fuse into one avatar at the beginning of the third part, to start following the user in an intrusive manner.

During the piece, the avatar(s) move closer to the spectator: the last part is staged as an interactive, following close up, that adds another layer of reality decomposition by introducing distorted facial expressions of the avatar. The piece ends with the avatar closing its eyes to disappear.

In the design of Mixed Reality scenarios, a distinction is made between global and local agency: when global agency is assigned to a user, their decisions cause permanent changes to the entire piece, while assigned local agency gives the possibility to influence individual factors that do not affect the overall event. For the sake of the dramaturgy within the dance performance, local agency competences of different sizes were assigned to the spectator to achieve a balance, so that despite influence, the piece remains perceptible in its entirety. Making use of local agency competences it is possible for the spectator to directly influence the piece through own movement and interaction, as the avatar dancers respond to proximity and distance, direction and angle of movement, and wind impulses. The interaction of the spectators with the avatars is designed to be „indirect“– triggered either by moving, inclining or tilting the device physically towards or away from the avatar, causing a dodging behavior, or by air flows, which are registered by the microphone and are causing distortions of the image as an effect on the wind streams.

Each of the three compositional parts begins with a set constellation of the avatars. During the evolution of each part, the avatar corresponds to each spectator interaction in real-time, using different patterns of reactions, depending on input, angle, and intensity of the approach. After dodging, the avatar continues to follow the choreography at a restored, reasonable distance at the new chosen place in virtual space until the end of that part. Each

subsequent part continues, after the dodged avatars have restored the needed constellation for that following part. As the piece evolves the avatar actively follows the visitor and moves very close in front of the camera image for the final scene. All reactions by the avatars to intended and unintended interactions of the visitor manifest in a uniquely arranged, non-repeatable performance each time the piece is screened. This way, the resulting performances represent a temporal-spatial, non-repeatable version of the digital dance piece *kin_* in the chosen exhibition space.



Figure 3: *kin_*: The AR Experience *kin_* in an Outdoor Setting

2 ARTIST INFORMATION

Performance artist and choreographer **Charlotte Triebus** researches the interface between dance, art, and technology with her ensemble New Human Body Society. See also www.charlottetriebus.com.

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